

Log1: May 25th 2026

1. Objective of This Session

The goal of this session was to verify whether the department GPU servers are capable of supporting:

- remote robotics development,
- ROS 2 + Gazebo simulation,
- GPU-accelerated workloads,
- GUI visualization over SSH,
- and containerized robotics environments.

This was the infrastructure validation phase before starting the actual robotics simulation stack.

2. Servers Evaluated

The following department GPU servers were initially considered:

- Ada
- Ampere
- Turing
- Kepler

Actual Outcome

Ada

Attempted SSH access to Ada failed.

Therefore:

- Ada was NOT used further during this session.
- No environment validation was performed on Ada.

Possible causes:

- temporary downtime,
- networking issue,
- SSH issue,
- or server-side restriction.

This will be revisited later.

Ampere

All subsequent testing and validation were successfully performed on:

```
ampere.ce.pdn.ac.lk
```

Ampere therefore became the primary validated server environment for the project at this stage.

3. Server Suitability Assessment

Based on the available server specifications and successful connectivity testing:

Server	Status	Assessment
Ada	SSH failed	Pending investigation
Ampere	Working	Primary validated environment
Turing	Not tested yet	Potential alternative
Kepler	Not recommended	Outdated hardware/software stack

Reasoning:

- Ampere contains modern RTX A6000 GPUs suitable for robotics simulation and future CUDA workloads.
- Kepler uses older Tesla K40 hardware with major CUDA compatibility limitations.

4. Local Machine Setup (Windows 11)

The following tools were installed/configured on the Windows laptop:

Tool	Purpose
MobaXterm	SSH + X11 forwarding
Windows 11	Client OS
SSH/X11	Remote Linux GUI access

This established the remote robotics workflow architecture:

```

Windows 11
  ↓ SSH/X11
Ampere Server
  ↓
ROS 2 + Gazebo + RViz

```

No Ubuntu dual boot or WSL installation was required.

5. SSH Connectivity Test (Ampere)

Command Used

```

[Inside MobaXterm -> Session -> Remote host:tesla.ce.pdn.ac.lk & Username:e21283] -> OK]
ssh e21283@tesla.ce.pdn.ac.lk
[now inside tesla, run the following command:]
ssh -X e21283@ampere.ce.pdn.ac.lk
[Now you will be inside ampere]

```

Result

SSH login succeeded successfully.

Conclusion

Verified:

- LDAP credentials functional
- Remote login access enabled
- Server reachable from current network

- X11 forwarding channel established

This confirmed that remote robotics development from Windows is feasible.

6. X11 / GUI Forwarding Verification

Command Used

```
[inside ampere]  
xclock
```

Result

A graphical clock window successfully opened on the Windows machine.

Conclusion

This confirmed:

- X11 forwarding works correctly
- MobaXterm X server functioning properly
- Remote GUI applications can be displayed locally

Implications:

- Gazebo GUI can potentially run remotely
- RViz2 visualization is feasible
- ROS GUI tools are usable remotely

This was one of the most critical validations of the session.

7. ROS 2 Availability Check

Commands Used

```
ros2 --version
```

```
ls /opt/ros
```

Output

```
ros2: command not found
```

and

```
No such file or directory
```

Conclusion

ROS 2 is NOT installed system-wide on Ampere.

However:

- this is not considered a blocker,
 - because Singularity is available,
 - allowing isolated containerized ROS environments.
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8. Gazebo Availability Check

Commands Used

```
gazebo --version
```

and

```
gz sim --version
```

Result

Gazebo not found.

Conclusion

Gazebo is not installed system-wide on Ampere.

Again, this is manageable through:

- Singularity containers,
 - user-space robotics environments,
 - or local ROS installations inside containers.
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9. GPU Verification

Command Used

```
nvidia-smi
```

Result

Detected:

- NVIDIA RTX A6000 GPUs
- CUDA 12.4 drivers
- Active GPU utilization

Conclusion

GPU acceleration is fully functional on Ampere.

This validates:

- future Gazebo GPU rendering,
- CUDA workloads,
- SLAM workloads,
- future Gaussian-LIC experimentation.

This is more than sufficient hardware for the simulation phase.

10. Shared Server Observation

During GPU inspection:

- GPUs were already heavily utilized by other users.

Conclusion

Ampere is an actively shared research server.

Implications:

- resource usage must be considerate,
 - long-running heavy workloads should be managed responsibly,
 - GPU scheduling awareness will be important later.
-

11. Singularity Verification

Command Used

```
singularity --version
```

Output

```
singularity-ce version 3.9.7-bionic
```

Conclusion

Singularity is installed and operational on Ampere.

This is a major project enabler because it allows:

- containerized ROS 2 environments,
- Gazebo installations without sudo,
- isolated dependency management,
- reproducible robotics environments.

This eliminates the immediate need for admin-installed ROS packages.

12. Storage Inspection

Commands Used

```
df -h
```

```
ls /storage
```

Results

Detected:

- `/storage/scratch1`
- `/storage/scratch2`
- `/storage/scratch3`

Also observed:

- `/home` partition is effectively full.

Conclusion

All robotics project work must be done inside scratch storage.

Recommended workspace location:

```
/storage/scratch2/$USER/go2_project
```

Critical note:

- Do NOT build ROS workspaces inside `/home`.

13. OpenGL / DISPLAY Observation

Command Used

```
glxinfo | grep OpenGL
```


Result

X11 forwarding works, but currently OpenGL rendering is running on the CPU instead of the NVIDIA GPU.

Rendering stack

Gazebo/RViz

↓

Mesa software renderer (llvmpipe)

↓

CPU rendering

↓

X11 forwarded to Windows

Note

When using ssh -X, OpenGL often falls back to Mesa software rendering because GPU OpenGL contexts are difficult over remote X11. It's not a server failure, this is standard Linux remote rendering behaviour.

Not bad for now, only becomes a problem when Gazebo worlds become heavy, sensors increase, camera streams increase, Gaussian-LIC rendering starts, multiple LiDARs run simulatenously.

We have to clearly distinguish the problem here. GPU compute is WORKING. (we confirmed it through the command `nvidia-smi`). The problem is only remote OpenGL visualization acceleration. Both are separate systems (Computation and Visualization).

We can go with several methods to solve this:

1. Headless simulation — Run `gazebo -- headless` Then, visualize only RViz or record rosbags or stream selectively. This avoids heavy GUI rendering.
2. Virtual Desktop — Use TurboVNC, VirtualGL, NICE DCV, these allow true GPU-accelerated OpenGL remotely. But they usually require admin setup.

14. Overall Conclusions

This session successfully validated the core infrastructure required for the project on Ampere.

Confirmed Working

Capability	Status
SSH access	Verified
Remote GUI forwarding	Verified
GPU visibility	Verified
Scratch storage	Verified
Singularity container support	Verified

Missing but Manageable

Component	Status
ROS 2	Not installed
Gazebo	Not installed

These can be solved using Singularity containers.

15. Strategic Outcome

The project is now confirmed to be technically feasible on department servers.

The infrastructure uncertainty has effectively been removed.

The workflow going forward will be:

```
Windows 11
  ↓
SSH/X11 via MobaXterm
  ↓
Singularity Container on Ampere
  ↓
ROS 2 + Gazebo
  ↓
Go2 Simulation
```

16. Immediate Next Steps

Step 1

Create project workspace inside scratch storage.

Example:

```
mkdir -p /storage/scratch2/$USER/go2_project
```

Step 2

Build or pull a ROS 2 Humble Singularity container.

Target environment:

- Ubuntu 22.04
- ROS 2 Humble
- Gazebo
- RViz2

Step 3

Validate robotics stack using TurtleBot3.

Goal:

- Launch Gazebo
- Spawn TurtleBot3
- Verify LiDAR topic publishing

This will serve as the first functional robotics validation milestone.

17. Key Technical Insights Learned

1. Remote robotics simulation from Windows is practical and stable.
2. GUI forwarding is functional and usable.
3. Containerization is the correct deployment strategy.
4. Scratch storage usage is mandatory.
5. No admin intervention is currently required.

6. Ampere is fully capable of supporting the simulation phase of the project.
 7. Ada availability must be investigated separately later.
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